A+ LAB SERIES V5

Lab 8: Windows Users and Groups

Document Version: 2026-01-02

Material in this Lab Aligns to the Following	
CompTIA A+ (220-1201) Exam Objectives	2.7: Compare and contrast internet connection types, network types, and their characteristics.
CompTIA A+ (220-1202) Exam Objectives	1.3: Compare and contrast basic features of Microsoft Windows editions. 1.6: Given a scenario, configure Microsoft Windows settings. 2.1: Summarize various security measures and their purposes. 2.2: Given a scenario, configure and apply basic Microsoft Windows OS security settings.

Contents

Introduction

One of the most important but often neglected activities a PC technician needs to perform is setting up user accounts. All major operating systems today (Windows, macOS, and Linux) can create user accounts with different levels of privilege and access to system resources.

One reason to create multiple users on a local computer is that multiple users will be using the same computer. In this way, each user can keep their data separate (and secure) from other users, and have different configurations and resource access settings.

The other reason is security and protection. This is to make sure that users have the access they need to get their work done, but no more. This refers to *the Principle of Least Access*. A logged-in user with administrative access is a disaster waiting to happen. Critical system files can be deleted, configuration options can be corrupted, and malware that has crept into the OS can be used to grant attackers remote access.

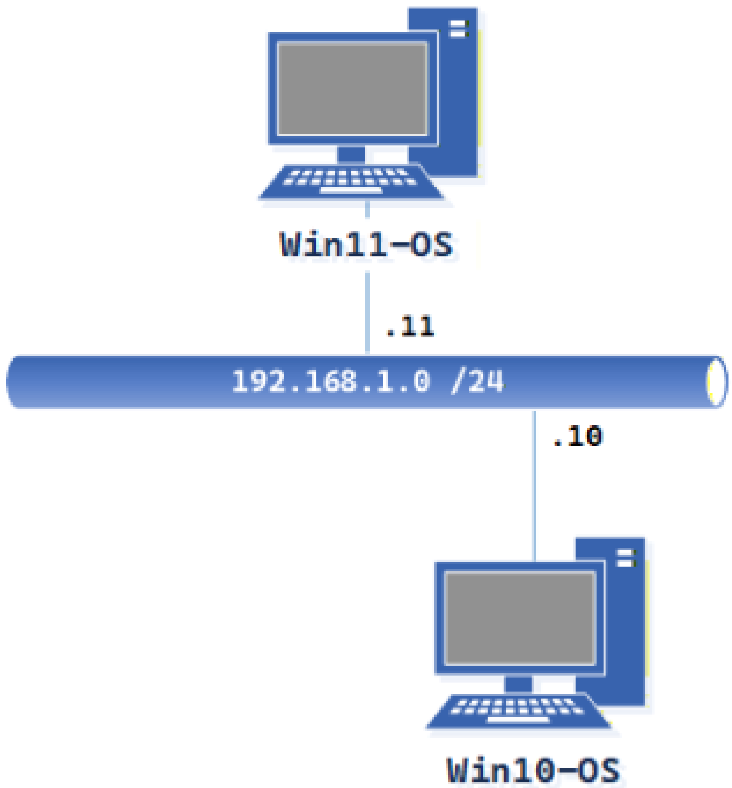
Even experienced users do not log in to an administrative account; instead, they use a standard, non-administrative account. All operating systems have a security feature built-in that requires an administrative user account and password when needed. In this way, a user can perform administrative tasks as needed and be alerted when one is requested (in case *someone or something else* requests it).

Objective

In this lab, we will perform tasks to create and manage user accounts and groups in Windows 10 and Windows 11, and review key concepts of Windows user and group management.

- Create and manage both administrative and standard user accounts, including passwords in Windows 10 and Windows 11.
- Test access between administrative and standard user accounts in both Windows 10 and Windows 11.
- Manage local group policies for account management.
- Use Computer Management to add and manage local users and groups.
- Use the Local Group Policy Editor to manage the Password Policy on Windows 10 and Windows 11.
- Install and run RSAT on Windows 10.
- Locate Administrative Tools to utilize Active Directory Users and Computers.

Lab Topology



Lab Settings

The information in the table below is needed to complete the lab. The task sections below provide details on how to use this information.

Virtual Machine	IP Address	Account (if needed)	Password (if needed)
Win10-OS	192.168.1.10 / 24	`sysadmin`	`NDGlabpass123!`
Win11-OS	192.168.1.11 / 24	`sysadmin`	`NDGlabpass123!`

1 Creating and Managing User Accounts

The next three sections will allow you to create and manage user accounts in *Windows 10* and *Windows 11*.

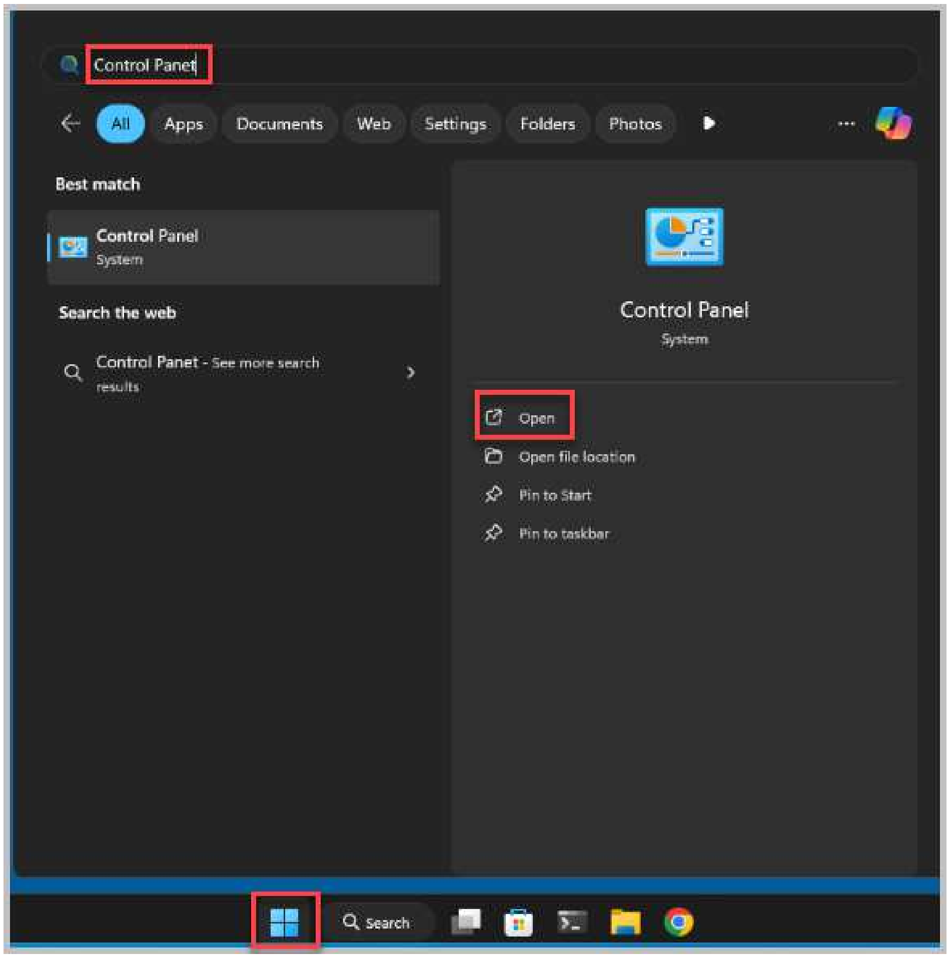
There are two primary ways to create users in *Windows*: the User Accounts applet in the Control Panel and *Local Users and Groups* in *Computer Management*. Each has its pros and cons, so let's create two users, each using a different tool.

1.1 Adding and Managing User Accounts in Windows 11

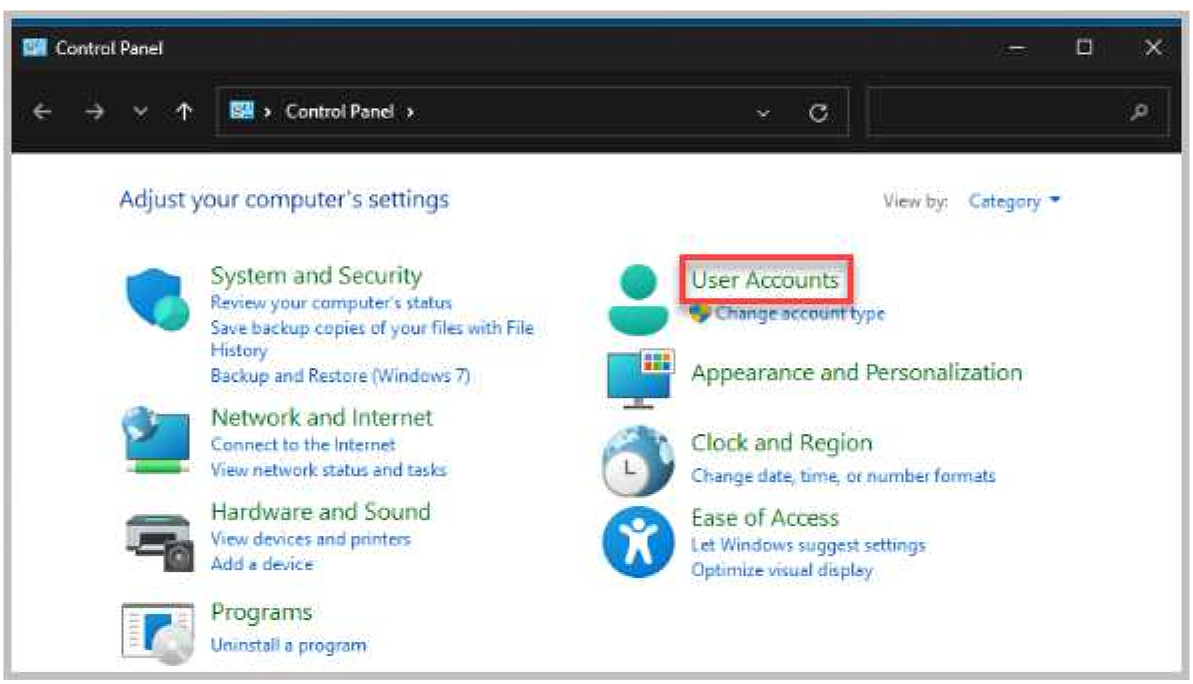
1. Open a console connection to the **Win11-OS** system.

> To launch the console for a lab device, you can click on the respective tab in the navigation bar at the top or navigate to the *Topology* section and click the lab device's corresponding image.

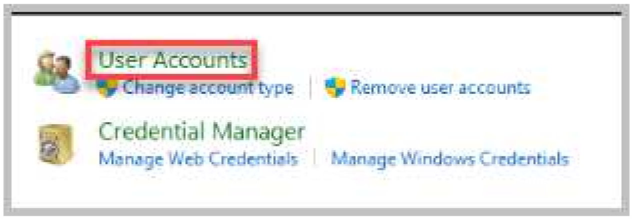
2. Click the **Start** button and type **Control Panel** in the *Search for Apps, Settings, and Documents* textbox. Select **Open.**



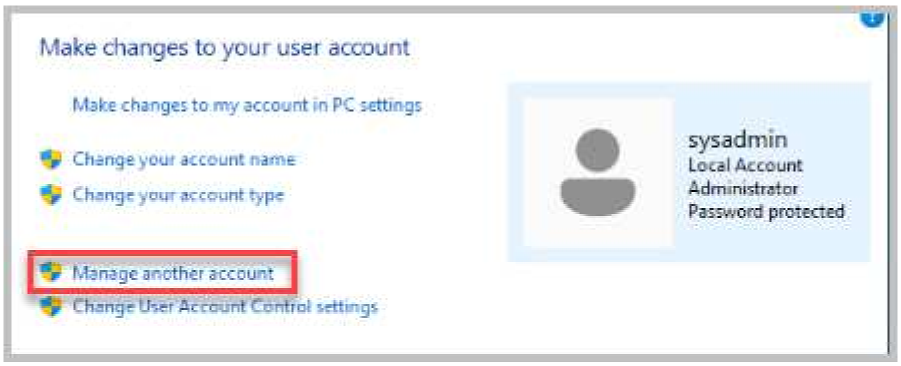
3. In the *Control Panel* window, click on **User Accounts**.



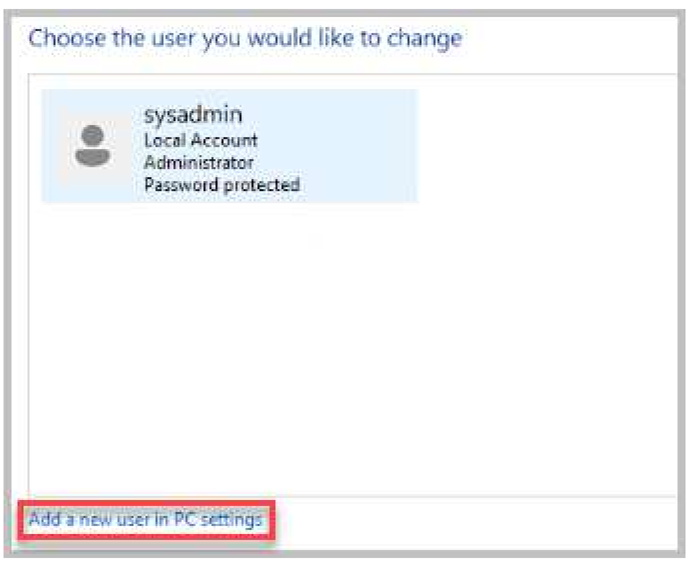
4. On the next screen, click **User Accounts**.



5. Click on **Manage another account**.

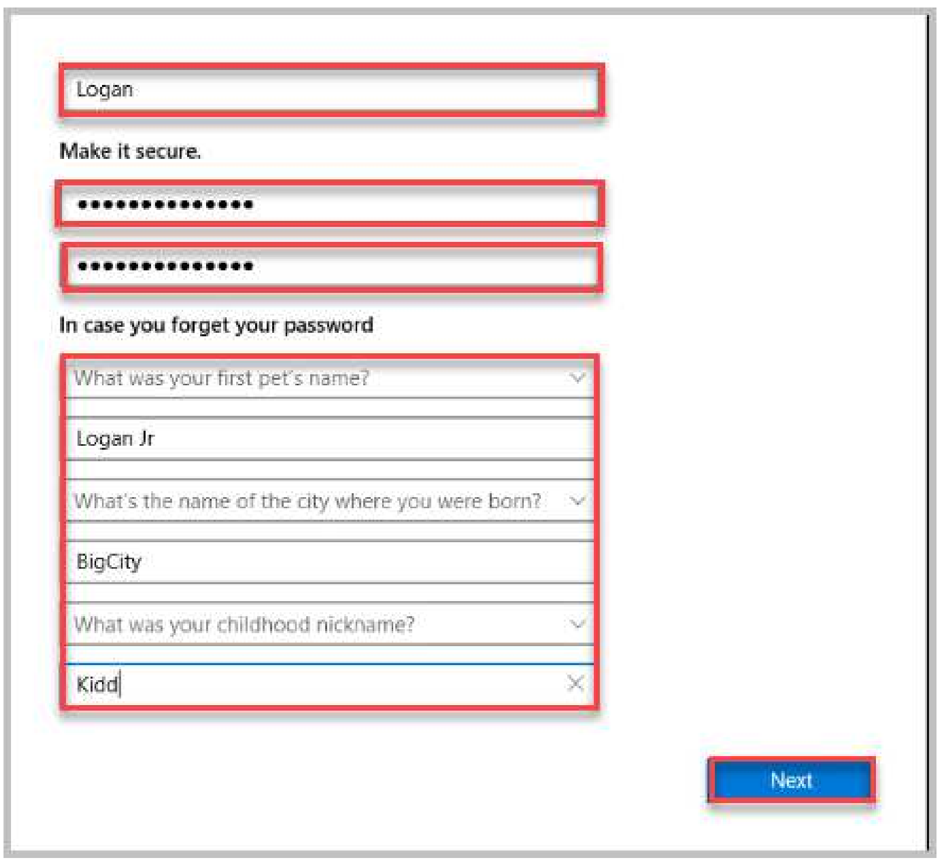


6. Click on **Add a new user in PC settings**.



7. Click on **Add Account.**

8. The next page allows you to create a user for the Win11-OS System. In the Who's going to use this PC textbox, type in **Logan.** Enter the password `NDGlabpass123!` and enter the same password again in the Re-enter password textbox. Next, you will need to choose 3 separate security questions and provide a unique answer to each. I chose random questions with generic answers. Security questions provide an additional layer of protection for your account if you forget your password. Select **Next.**



When setting security questions, it's important to choose questions whose answers are not easily guessable by others or available on social media and to use answers that are memorable to you but not obvious. Remember, the effectiveness of security questions depends on how unique and secure the answers are.

Passwords keep unauthorized users out of the system and keep the information on the computer safe and secure. But a password only works if it is difficult to guess or crack.

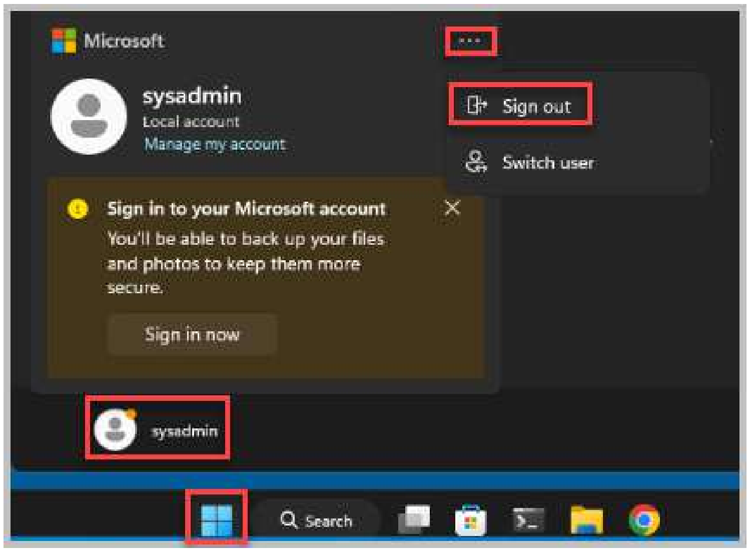
Good password rules:

- Security experts currently recommend a minimum of 8 characters.
- Use a combination of upper/lower case letters, numbers, and special characters.
- Do not use common words and NEVER use self-identifying information (spouse's, pet's, children's names, etc.).

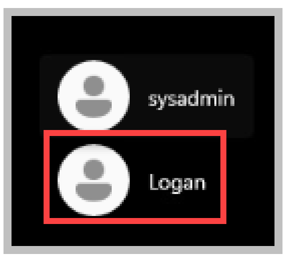
9. Notice that *Logan* is now added to the list of Other users for this machine. To make a couple of customizations to the account, click on the **Logan** account. The options are to change the account type or remove it if it is no longer needed.

10. Close the **Other Users** accounts window. Close the **Manage Accounts** window.

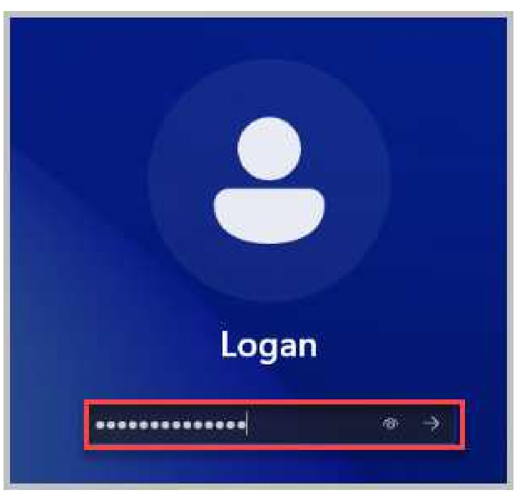
11. Click the **Start** button, then click the **sysadmin** account. Click the **three dots** at the top of the account window and click **Sign out.**



12. Now, log in to the newly created **Logan** account by clicking on the account name in the bottom left of the window.

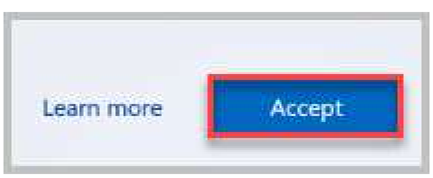


13. You will be asked to enter the Logan's password. Type `NDGlabpass123!` followed by pressing the **Enter** key to log in or selecting the right arrow next to the password you typed.



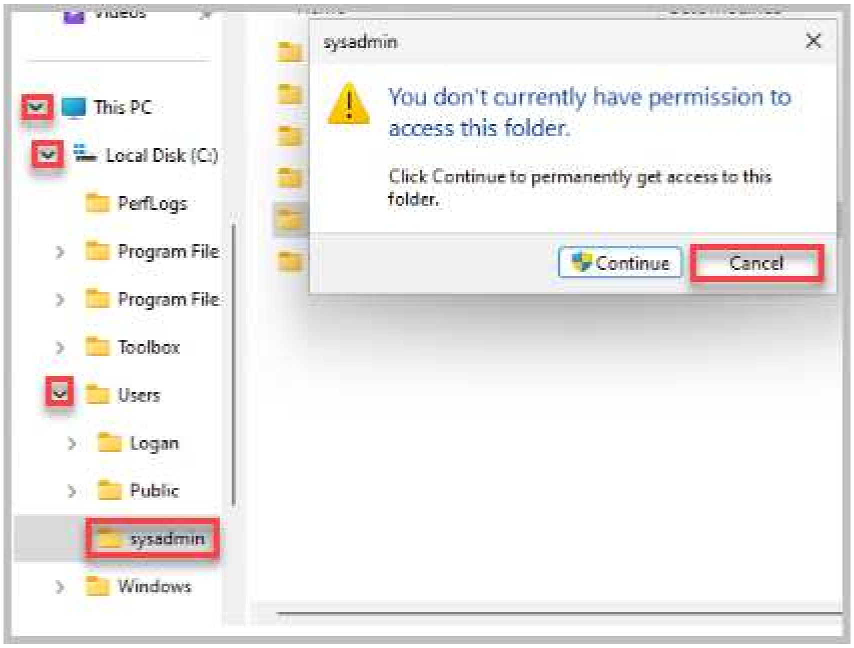
14. As *Logan* is a new user account for the Win11-OS system, you will need to complete the setup for the account. On the "Choose the privacy settings for your device" page, simply review the options presented. You can leave all settings at their default configurations for now. Once you've looked them over, click **Next** to proceed.

15. Click **Accept** to save your selections.

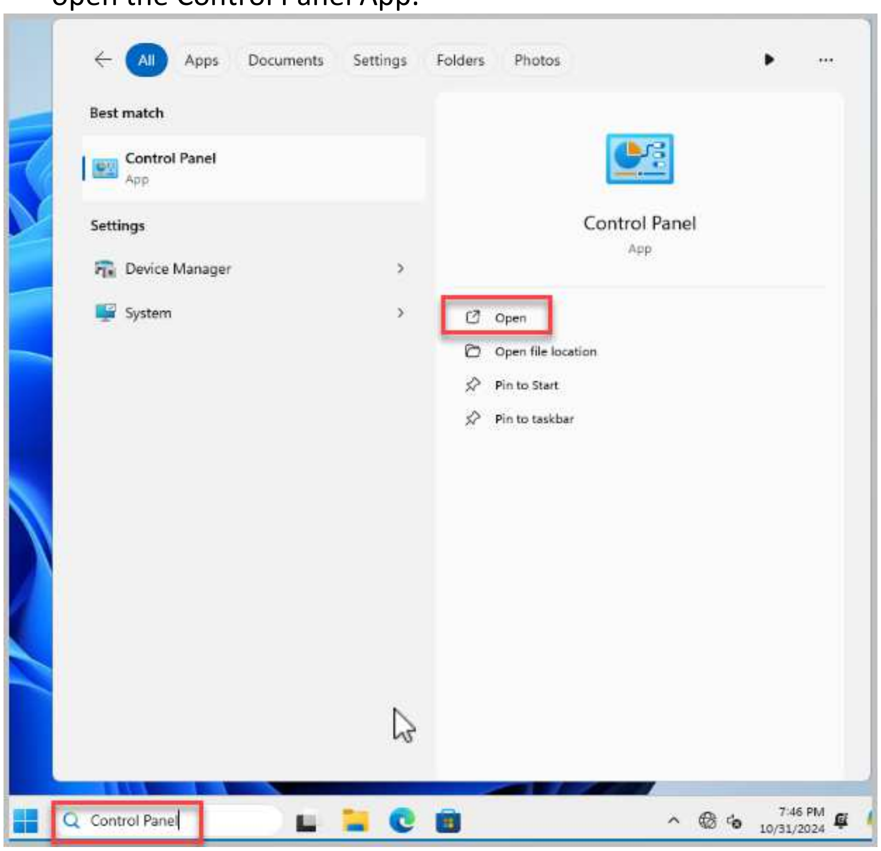


16. On the Win11-OS taskbar, click the **Windows Explorer** icon.

17. Using *Windows Explorer*, select the **pointer arrow** next to This PC. Select **Pointer Arrow** next to Local Disk (C:). Select the **Pointer Arrow** next to Users. Select the **sysadmin** folder icon. Notice the pop-up window that says you cannot access the folder because you do not have the proper credentials. Click the **Cancel** button to close the window.

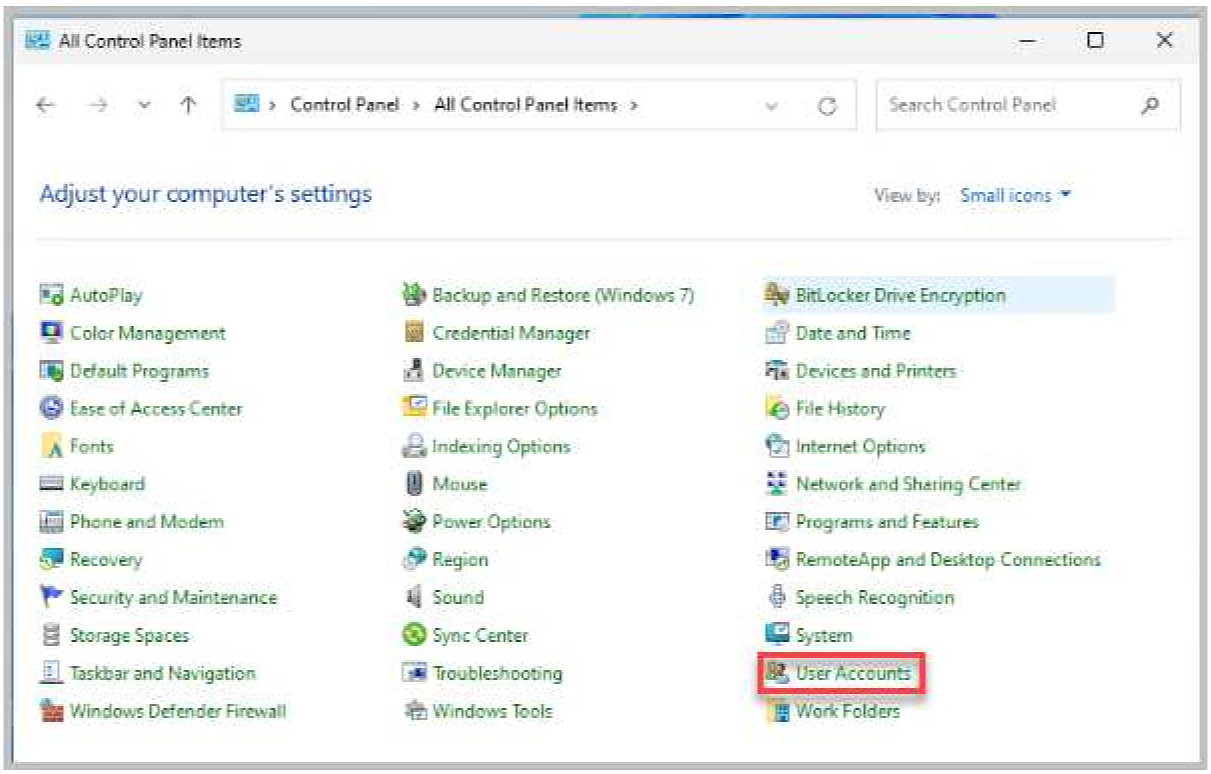


18. Click the **X** in the *File Explorer* window to close it.
19. On the taskbar of the Win11-OS system, type **Control Panel** in the Search textbox. Select **Open** to open the Control Panel App.

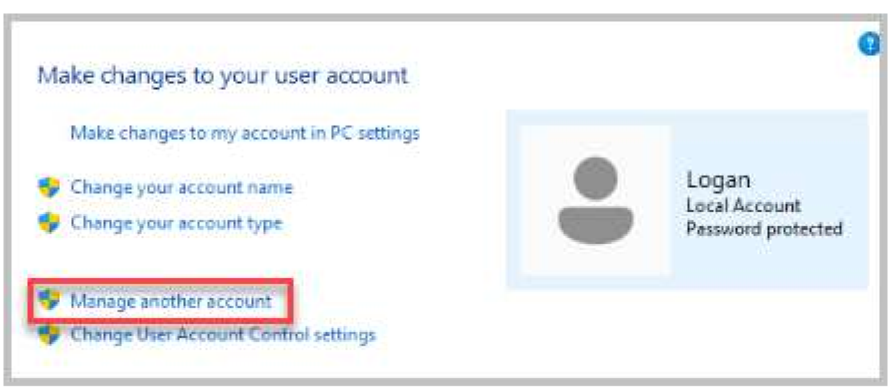


20. Change the *View by:* to **Small icons**.

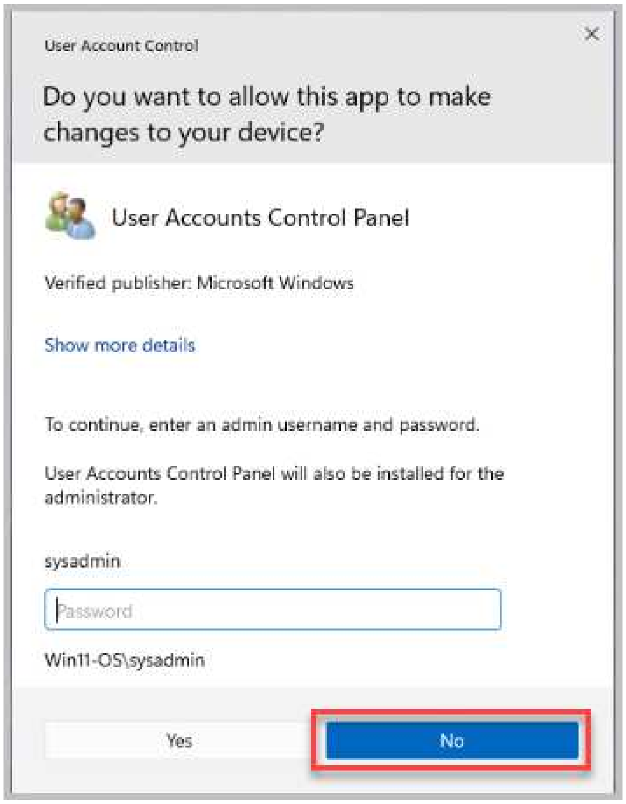
21. Click on **User Accounts**.



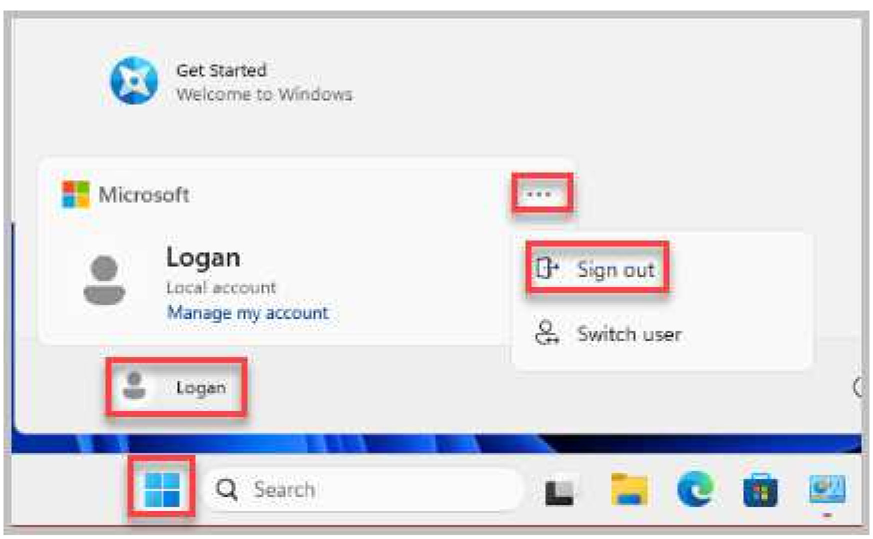
22. On the *User Accounts* window, click on **Manage another account**.



23. This time, you get a *User Account Control* message prompting you to enter the administrator's password. Standard users cannot create or manage other user accounts. If you entered the administrator's password here, you would be granted access. This is how administrators use their standard accounts, but still can perform administrative functions safely. For now, click the **No** button.



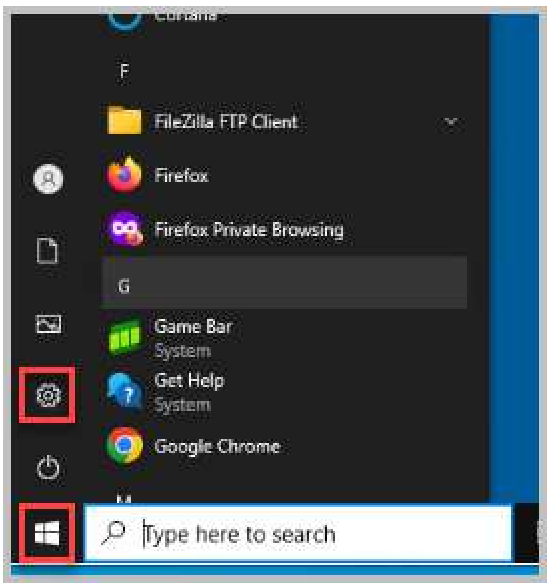
24. Click the **Start** button, then click the **Logan** account. Click the **three dots** at the top of the account window and click **Sign out.**



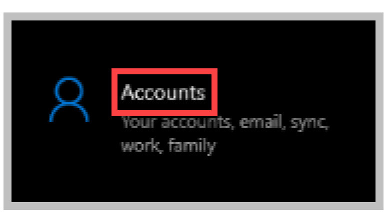
1.2 Adding and Managing User Accounts in Windows 10

1. Open a console connection to the **Win10-OS** system.

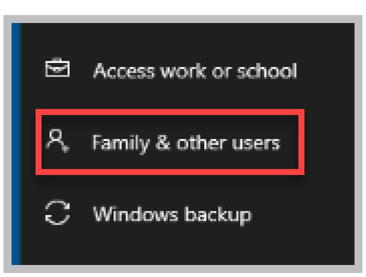
2. Adding users in *Windows 10* can be started from the *Control Panel*, but the applet will take you to *PC Settings,* which is where most configuration settings are made. Click on the **Start** button and then click on the **Settings** icon.



3. In the *Windows Settings*, click on **Accounts**.



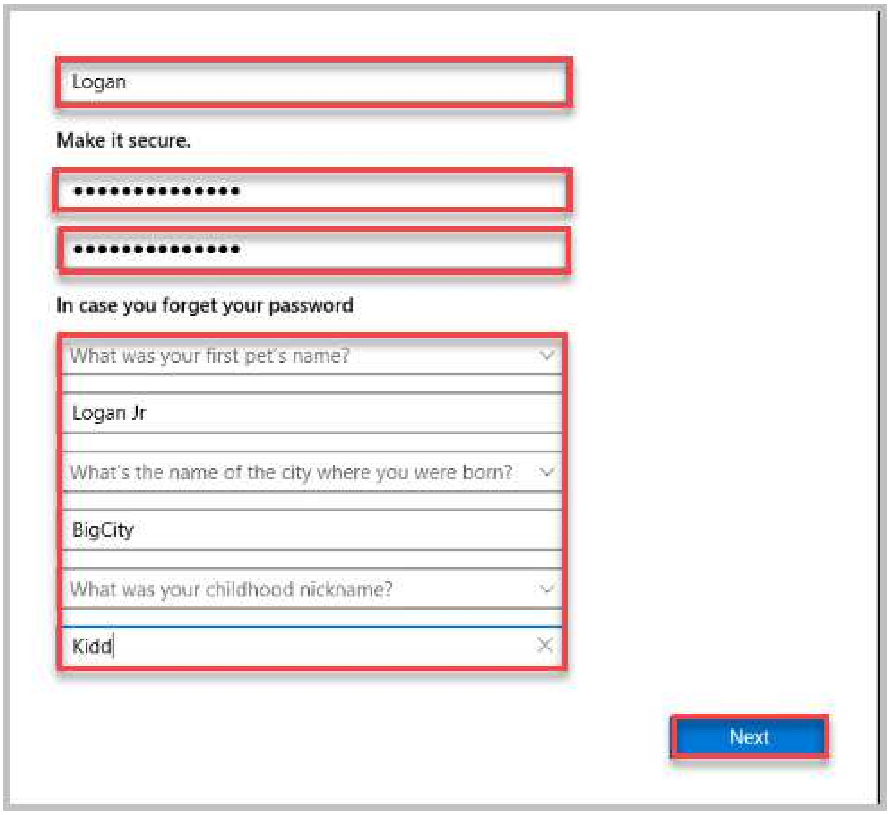
4. Click on **Family & other users** from the left pane.



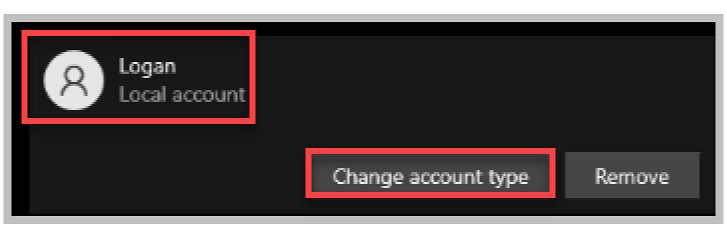
5. Underneath the *Other users* header, click on **Add someone else to this PC**.

6. Enter To add Logan as a user, enter the required information in the fields provided, then click on **Next** to continue.

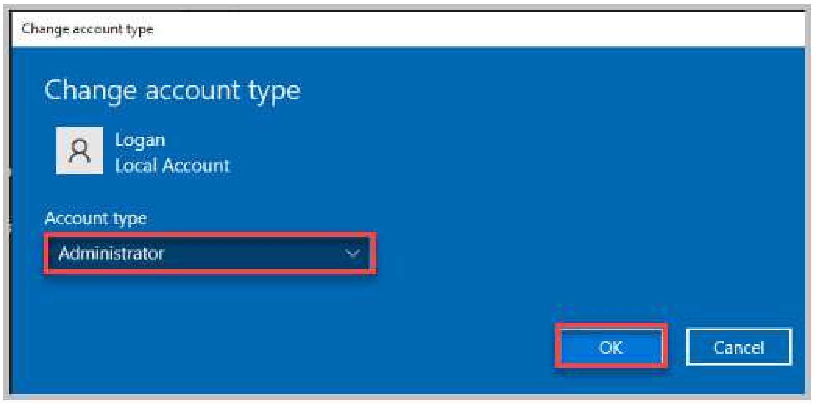
   a. *User name*: `Logan`
   b. *Password*: `NDGlabpass123!`
   c. *Security Questions*:
      1) What was your first pet's name? `Logan Jr`
      2) What was the name of the city where you were born? `BigCity`
      3) What was your childhood nickname? `Kidd`



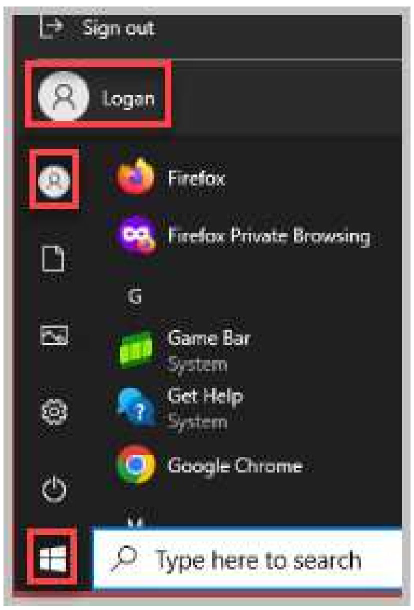
7. Notice the new user account is now created and that it is a local account. Just like *Windows 11,* a newly created user is automatically a *Local Account.* To make the *Logan* user account an administrator, click on **Logan**, then click the **Change Account Type** button.



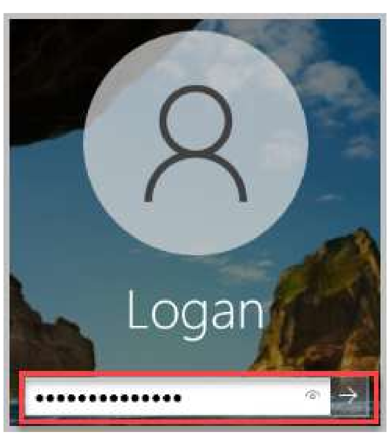
8. Under *Account type*, change it to **Administrator**. Click **OK.**



9. Close the Family & Other Users window. Now, change from using the *sysadmin* account and log in to the **Logan** account. Click the Start button, then click the **sysadmin** icon. Choose the **Logan** account to sign in.



10. You will be asked to enter the user's password, enter `NDGlabpass123!` followed by pressing the **Enter** key.



11. Notice the *Windows* operating system is setting up the user account for first-time use. Select the default settings for Choosing the privacy setting for your device by clicking **Accept.**

12. Log out of account *Logan.*

2 Using Local Users and Groups in Computer Management

Now, you will create a user using *Local Users and Groups* in *Computer Management*. This method allows you to control more of the information that is entered.

Please Note

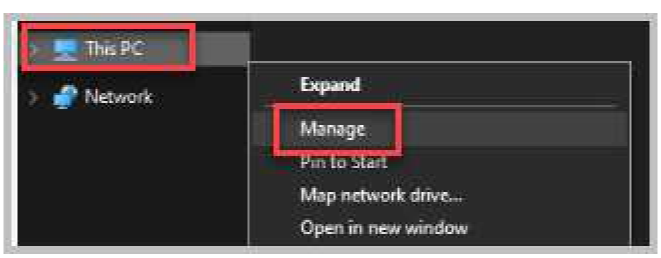
Creating a user via Local Users and Groups in Computer Management is similar in both Windows 10 and 11, offering detailed control over user account settings. While minor UI differences might exist between versions, the core process remains consistent, allowing for effective user management across these operating systems.

2.1 Adding and Managing User Accounts in Windows 10

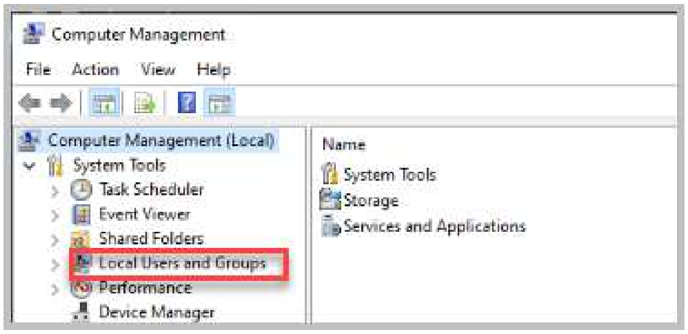
1. If not already, change focus to the *Win10-OS* system. Log in as the `sysadmin` account with `NDGlabpass123!` as the password.

2. In the taskbar, click **File Explorer.**

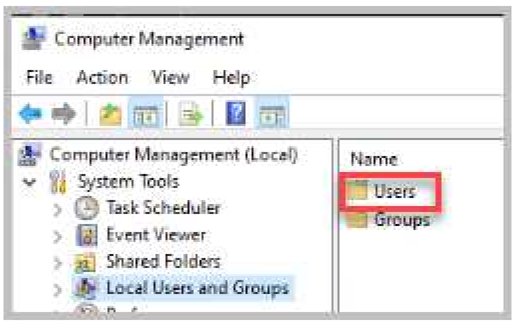
3. Right-click the **This PC** icon and click **Manage**.



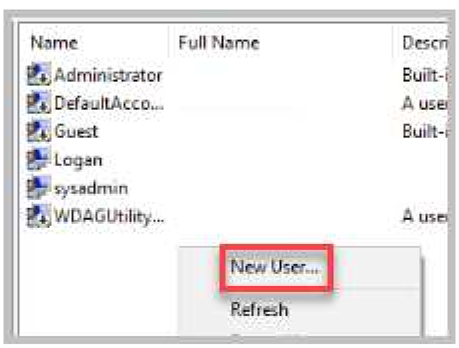
4. In the *Computer Management* window, click on **Local Users and Groups** from the left pane.



5. In the center pane, double-click on the **Users** folder, which will open the list of users.

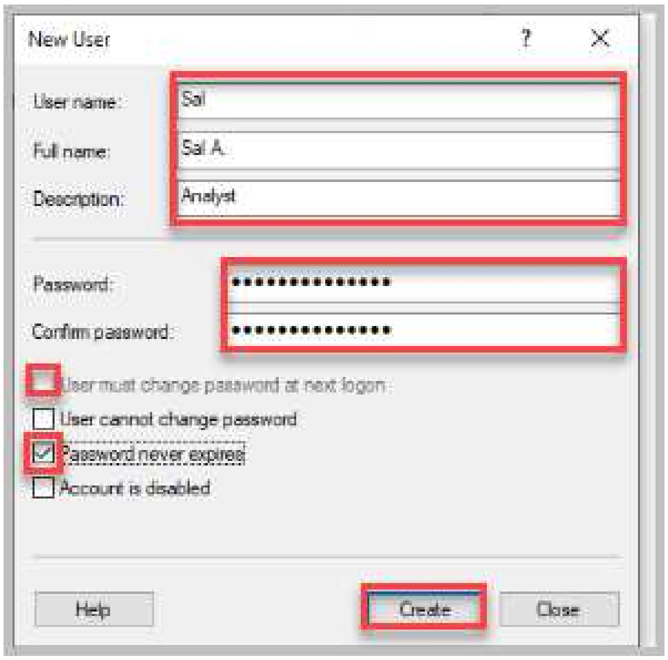


6. In the center pane, right-click anywhere within the whitespace and select **New User**.

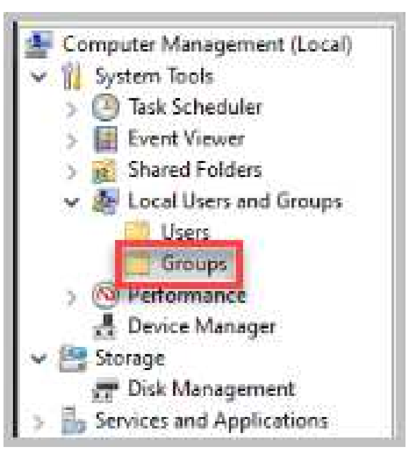


7. In the *New User* window, enter the following information:
   a. *User name*: `Sal`
   b. *Full name*: `Sal A.`
   c. *Description*: `Analyst`
   d. *Password*: `NDGlabpass123!`
   e. *Confirm Password:* `NDGlabpass123!`

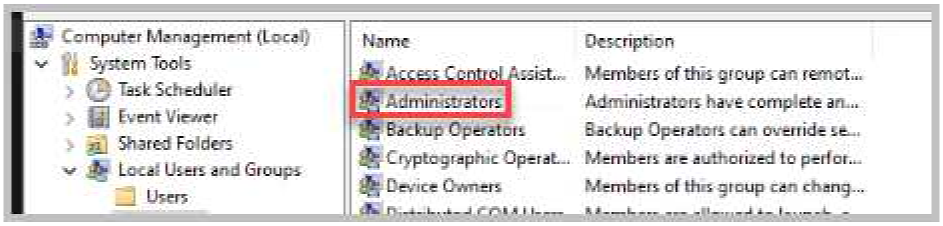
Uncheck the "User must change password at next logon" checkbox, then check the "Password never expires" checkbox. Click the **Create** button.



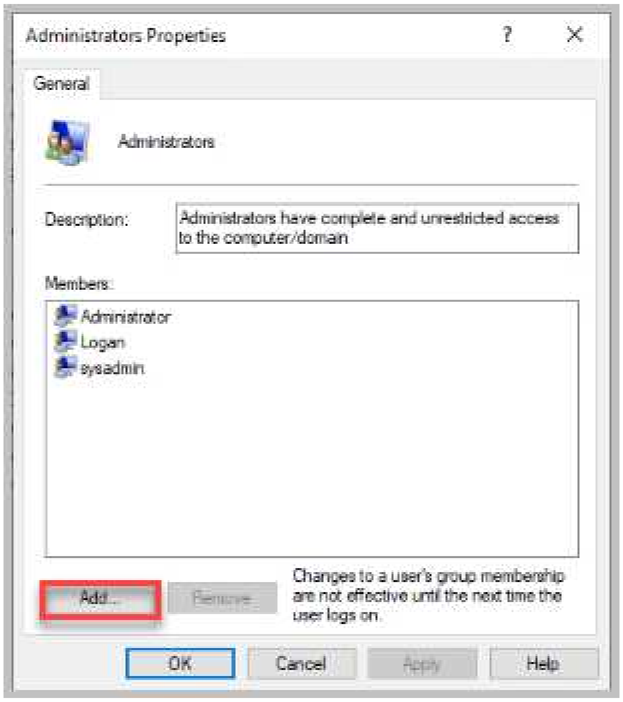
8. In the *New User* window, click the **Close** button to close the window.
9. By default, any new user created using the *Computer Management* tool is a *Standard User*. To grant the user administrative privileges, you must add them to the administrator group. Click Groups under Local Users and Groups in the left pane.



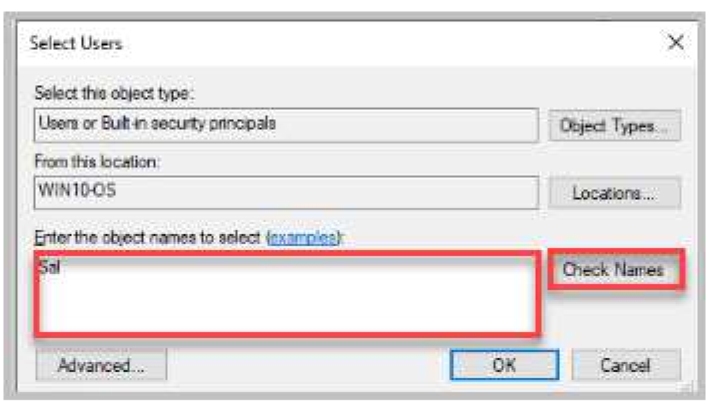
10. Double-click on the **Administrators** group located in the center pane.



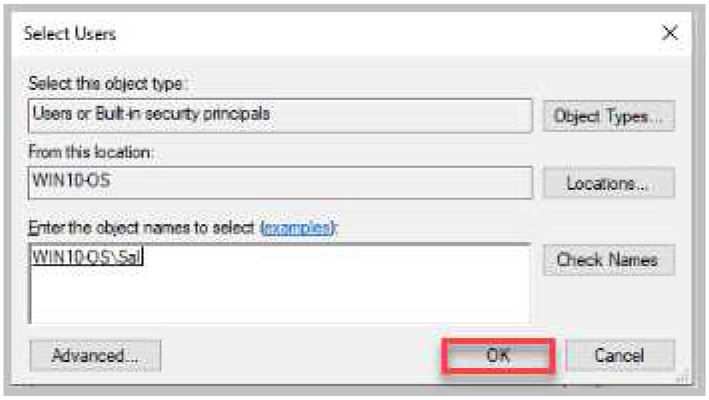
11. In the *Administrator Properties* window, click on the **Add** button.



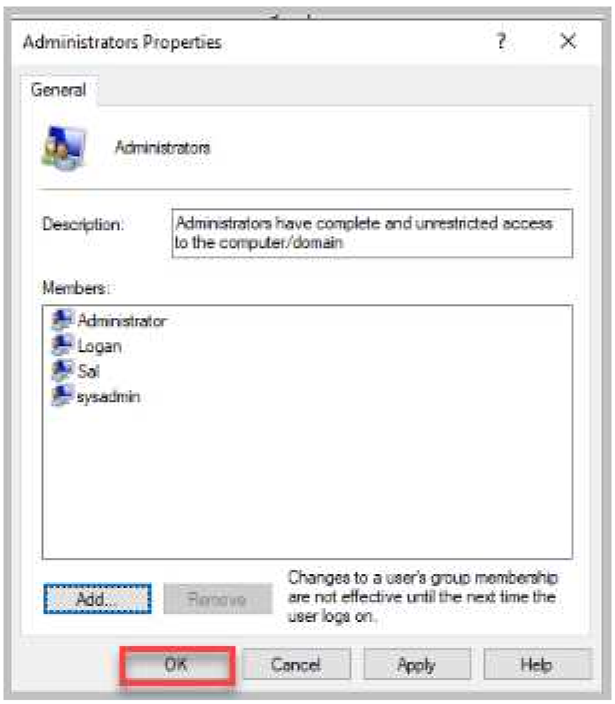
12. On the *Select Users* window, type `sal` into the *Enter the object names to select* and then click the **Check Names** button.



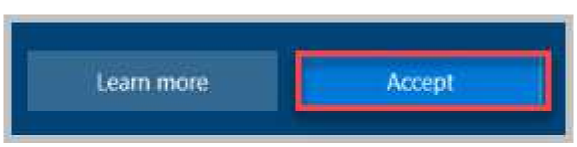
13. The username should now appear in the format *<COMPUTER>\<USER>*. If it doesn't, go back and check the username spelling and try again. If the username is found, click OK.



14. The *Administrator Properties* should now show *Sal* as a member of the *Administrators* group. Click **OK** to close the *properties* window and then close the *Computer Management* window.



15. Close the *Computer Management Window*. Close the *File Explorer Window*. Log out of the *sysadmin* account, then log back in as **Sal A.** Now that *Sal* is a member of the *Administrators* group, they can access other users' files and folders.
16. As this is a new account, a simple setup is required. Select **Accept to accept the default settings and choose your device's privacy settings**. Move on to the next activity.



3 Creating and Managing Local Group Policies

From a security perspective, a strong password alone is still not enough. One of the tasks that will fall onto a PC technician is to set up and enforce good password policies.

A good password policy should have the following characteristics:

- Passwords should be changed periodically
- Passwords should have a minimum length of 8 characters
- Passwords should be complex (a combination of upper/lower case, numbers, and special characters)
- Passwords should not be reusable

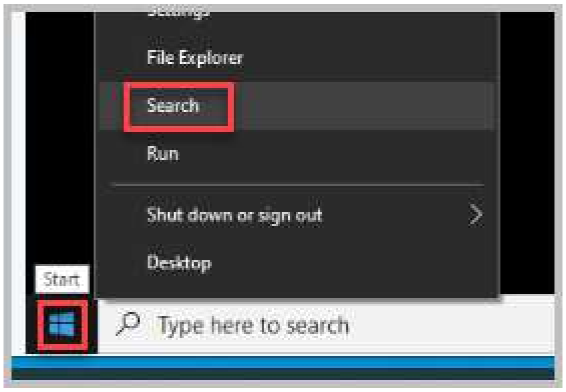
While these policies can be written down in a policy and procedures guide, it would be better if the operating system could enforce them. *Windows* provides a mechanism for enforcing these policies through the *Local Group Policy Editor* (a local version of *Windows Active Directory Group Policies*).

In this portion of the lab, we will use the *Local Group Policy Editor* to change the user account's password policies.

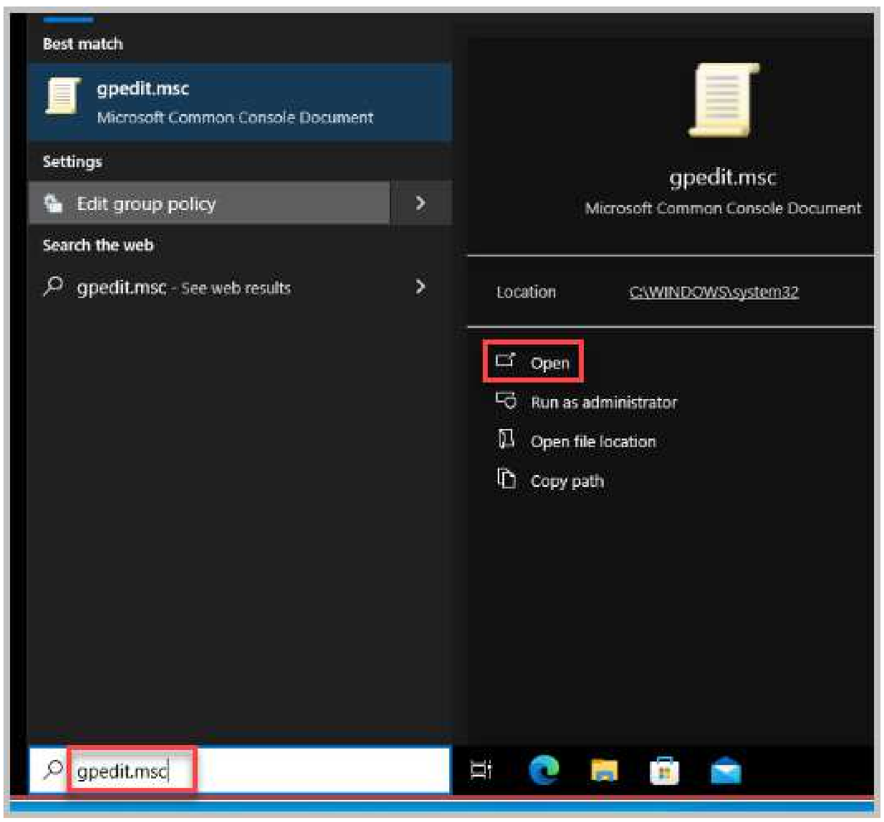
3.1 Using the Local Group Policy Editor in Windows 10

1. If not already, change focus to the **Win10-OS** machine. From the previous activity, you created an account, Sal A., which is part of the administrators group.

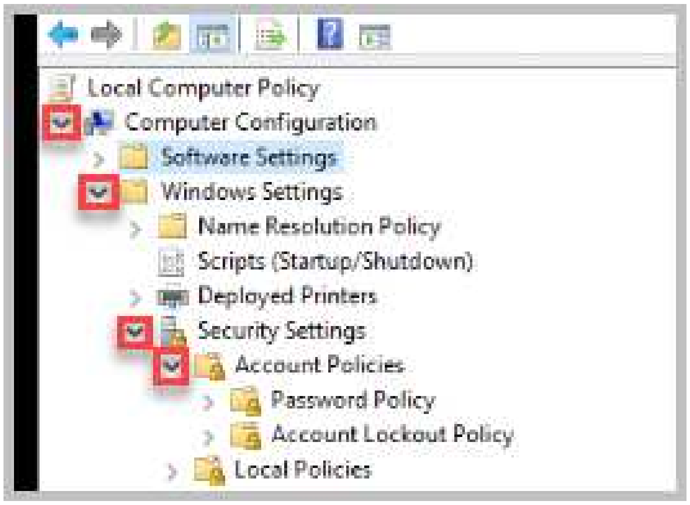
2. Right-click on the **Start** button and select **Search**.



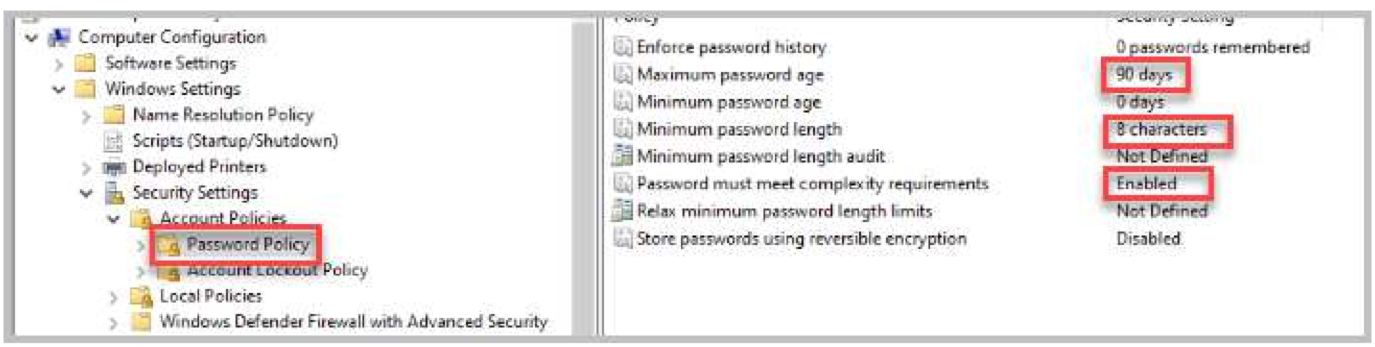
3. In the *Search* text field, type in `gpedit.msc` followed by clicking **Open.**



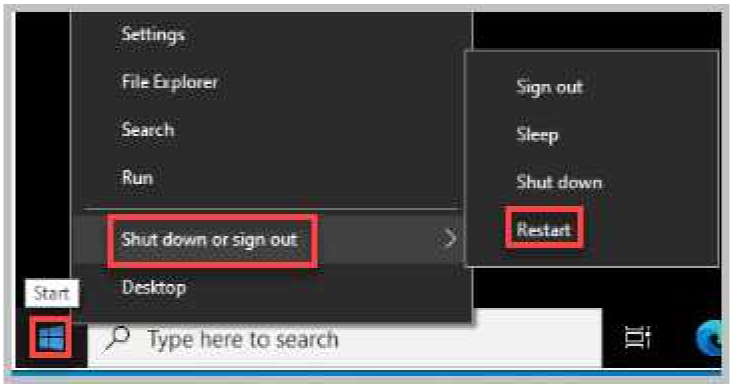
4. In the *Local Group Policy Editor*, click the arrow to the left of **Windows Settings** underneath *Computer Configuration*, then **Security Settings**, and finally, **Account Policies** to expand their respective views.



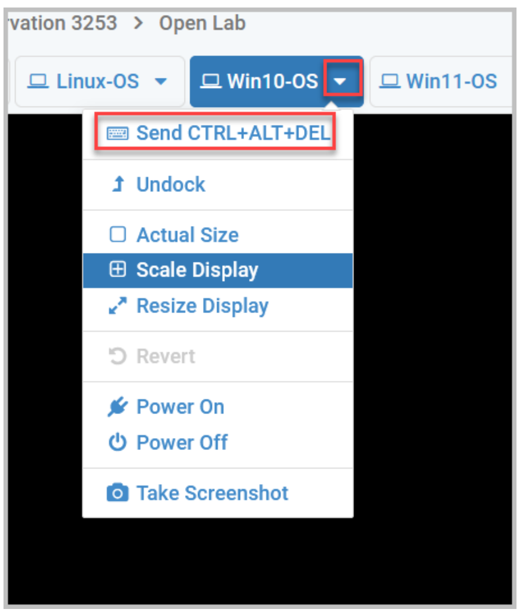
5. Under *Account Policies,* click **Password Policy** to show all password policy options on the right. Make the following changes by double-clicking on each policy that is listed in the right pane, after making the change select *Ok* to close the current window. Continue to change the remaining items as listed below:
   a. *Maximum password age*: `90 days`
   b. *Minimum password length*: `8 characters`
   c. *Password must meet complexity requirements*: **Enabled**



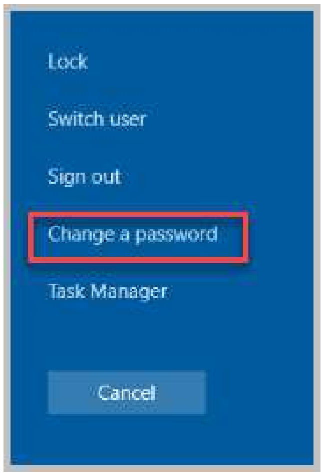
6. Once the password policy is updated, close the *Local Group Policy Editor w*indow. Reboot the Win10-OS computer to make the changes effective. Right-click the **Start** button and select **Shut down or sign out > Restart**.



7. After rebooting, the Win10-OS will automatically log in with the *sysadmin* account.
8. Sign out of the *sysadmin* account and use the password `NDGlabpass123!` to sign into the *Logan* account.
9. Once logged in, you need to enter a **CTRL+ALT+DEL** sequence. In the *NETLAB+* navigation bar, click the arrow next to the **Win10-OS** virtual machine tab, then click on **Send CTRL+ALT+DEL**.



10. Click on **Change a password**.



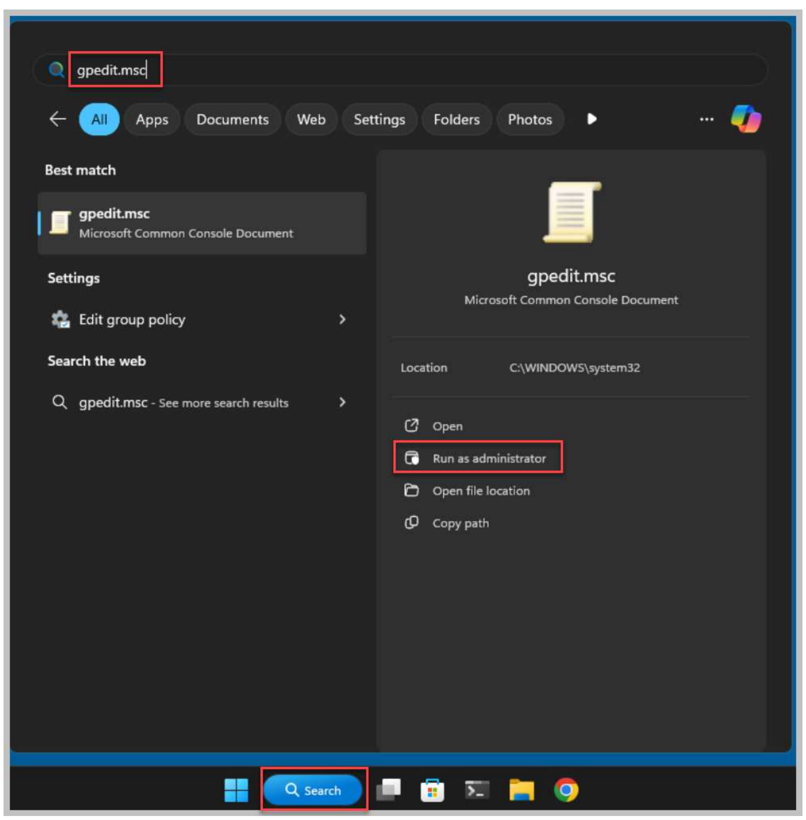
11. Change the account to Logan *and e*nter **`NDGlabpass123!`** as the old password. For the new password, type **`Pass1`**. Confirm the password, then click the arrow or press **Enter**.

12. Notice the error message stating that the password does not meet the current policy requirements. Click **OK**. Now try changing the password again, but use `NDGlabpass1234!` as the new password (it has fifteen characters). Click the **Arrow** to confirm the change.

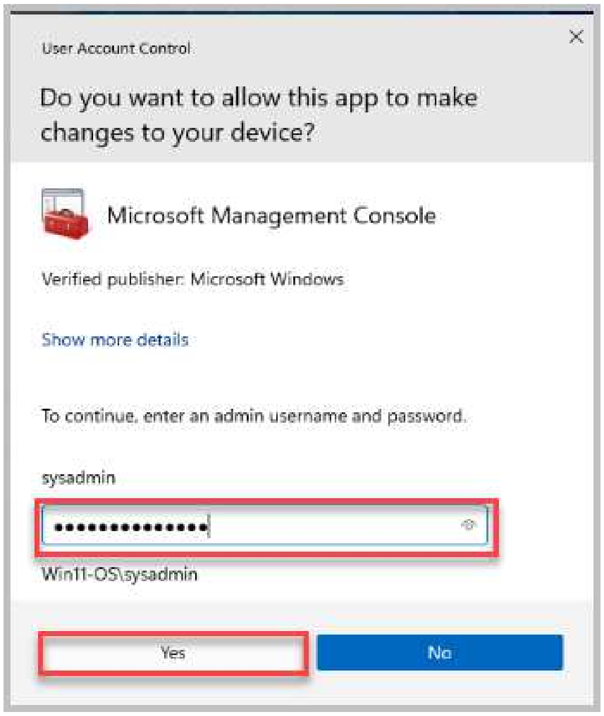
13. Log out and log back in as **Logan** to confirm the new password has been applied successfully. After successfully verifying the Logan account, log out and proceed to the next activity.

3.2 Using the Local Group Policy Editor in Windows 11

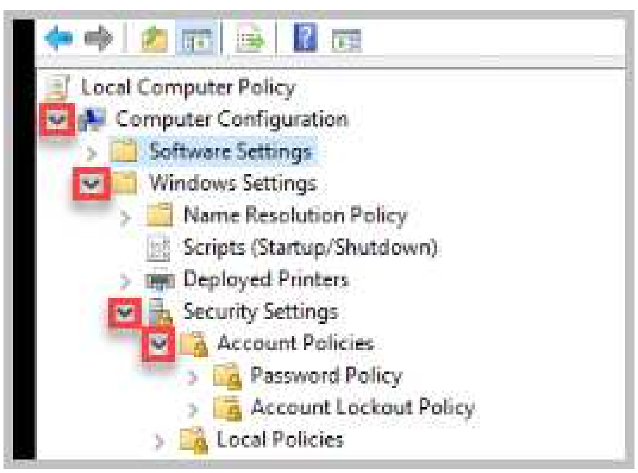
1. Change focus to the **Win11-OS** machine. Log in with the account *sysadmin* and the password `NDGlabpass123!`. The *sysadmin* account is part of the administrator's group.
2. Select **Search,** and in the *Search* text field, type in `gpedit.msc` followed by clicking **Run as administrator.**



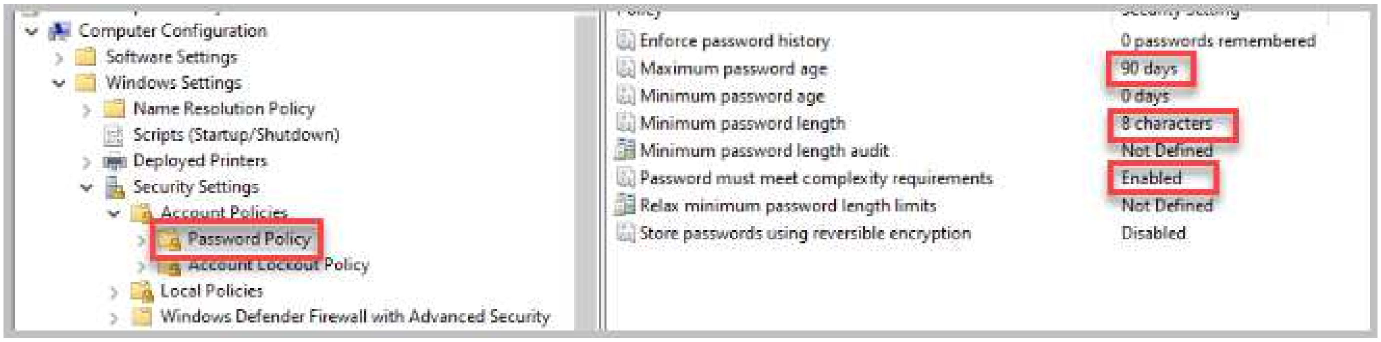
3. If prompted, in the *User Account Control* pop-up notification, enter the *sysadmin password* `NDGlabpass123!` and select **Yes.**



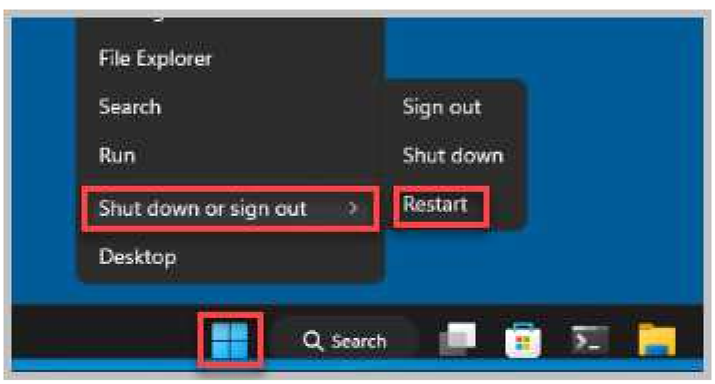
4. In the *Local Group Policy Editor*, click the arrow to the left of **Windows Settings** underneath *Computer Configuration*, then **Security Settings**, and finally, **Account Policies** to expand their respective views.



5. Under *Account Policies,* click **Password Policy** to show all password policy options on the right. Make the following changes by double-clicking each policy listed in the right pane. After making the change, select *Ok* to close the current window. Continue to change the remaining items as listed below:
   Maximum password age: `90 days`
   Minimum password length: `8 characters`
   Password must meet complexity requirements: **Enabled**

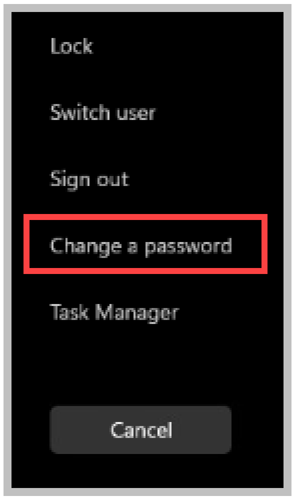


6. Once the password policy is updated, close the *Local Group Policy Editor* window. Reboot the Win11-OS computer to make the changes effective. Right-click the **Start** button and select **Shut down or sign out > Restart**.

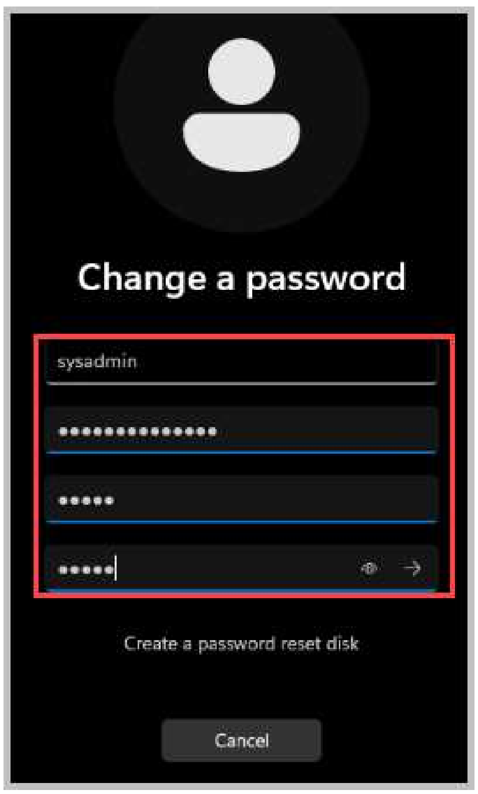


7. After rebooting, the Win11-OS system will automatically log in with the *sysadmin* account.
8. Once logged in, you need to enter a **CTRL+ALT+DEL** sequence. In the *NETLAB+* navigation bar, click the arrow next to the **Win11-OS** virtual machine tab, then click on **Send CTRL+ALT+DEL**.

9. Click on **Change a password**.

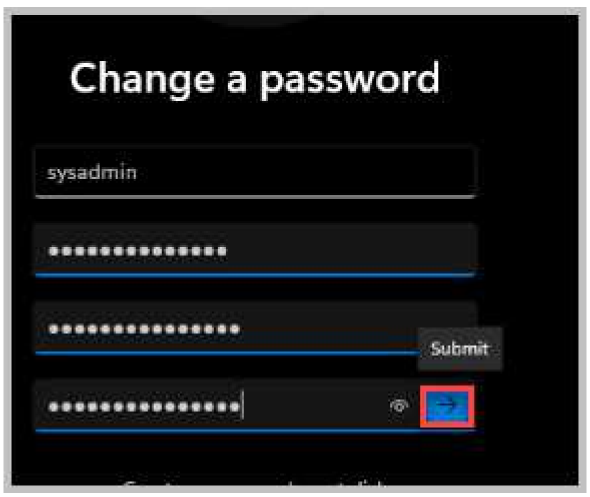


10. Make sure the account is listed as sysadmin *and e*nter `NDGlabpass123!` as the old password. For the new password, type `Pass1`. Confirm the password, then click the arrow or press **Enter**.

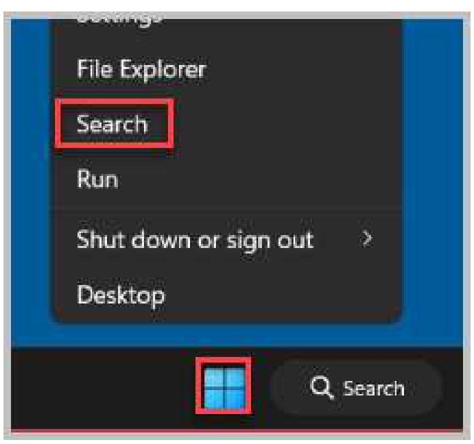


11. Notice the error message stating that the password does not meet the current policy requirements. Click **OK**.

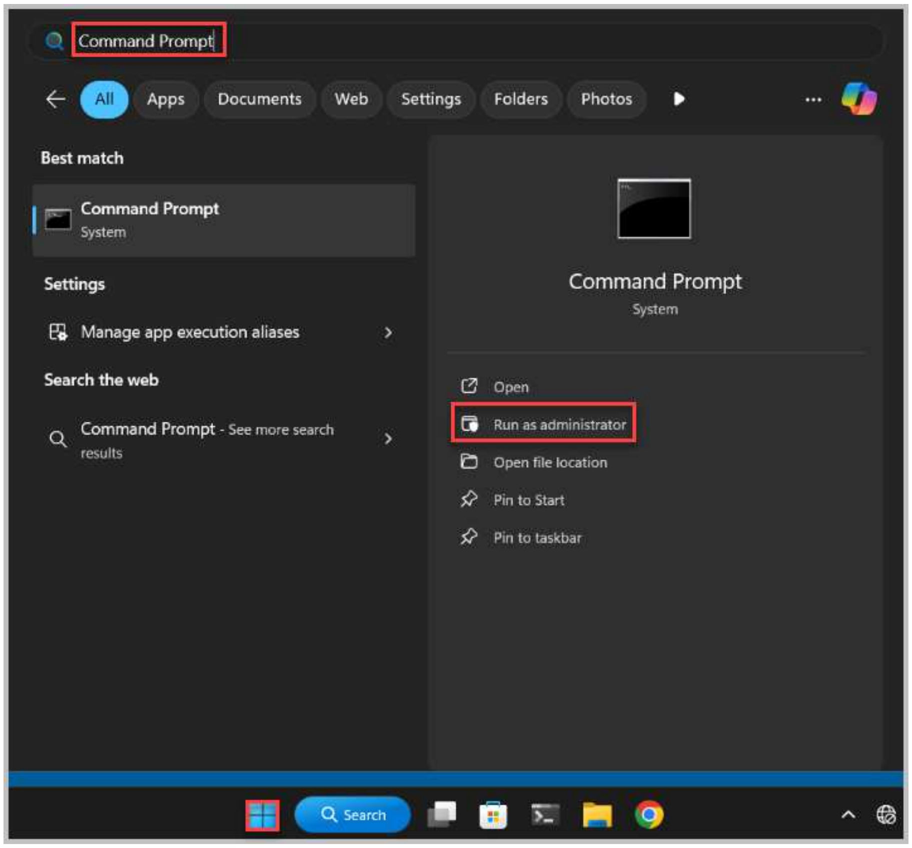
12. Now, try again to change the password but use `NDGlabpass1234!` as the new password (it has fifteen characters) and select the **pointer arrow** to confirm the change. Select **OK** to acknowledge the password has been changed.



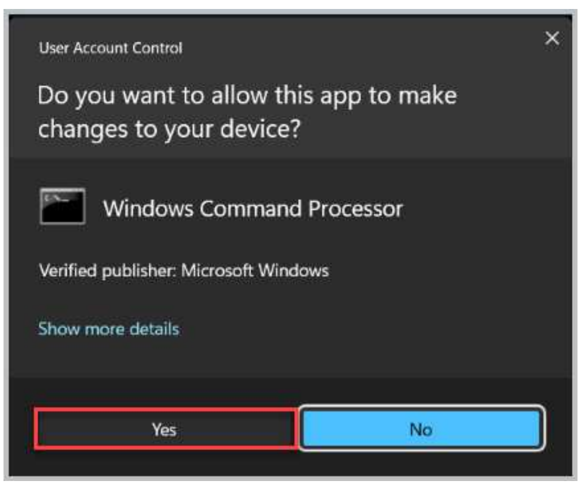
13. Log out and log back in as **sysadmin** to confirm the new password has been applied successfully.
14. Right-click the **Start** button and select **Search.**



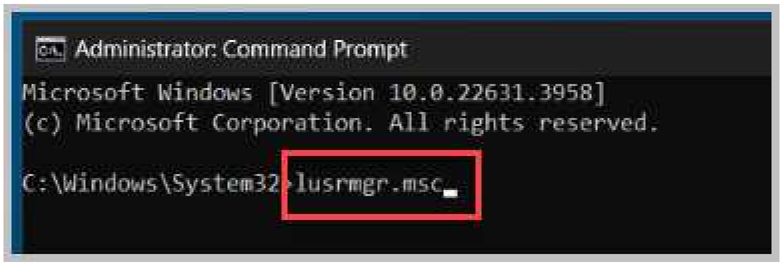
15. In the search box, type **Command Prompt** and select **Run as administrator.**



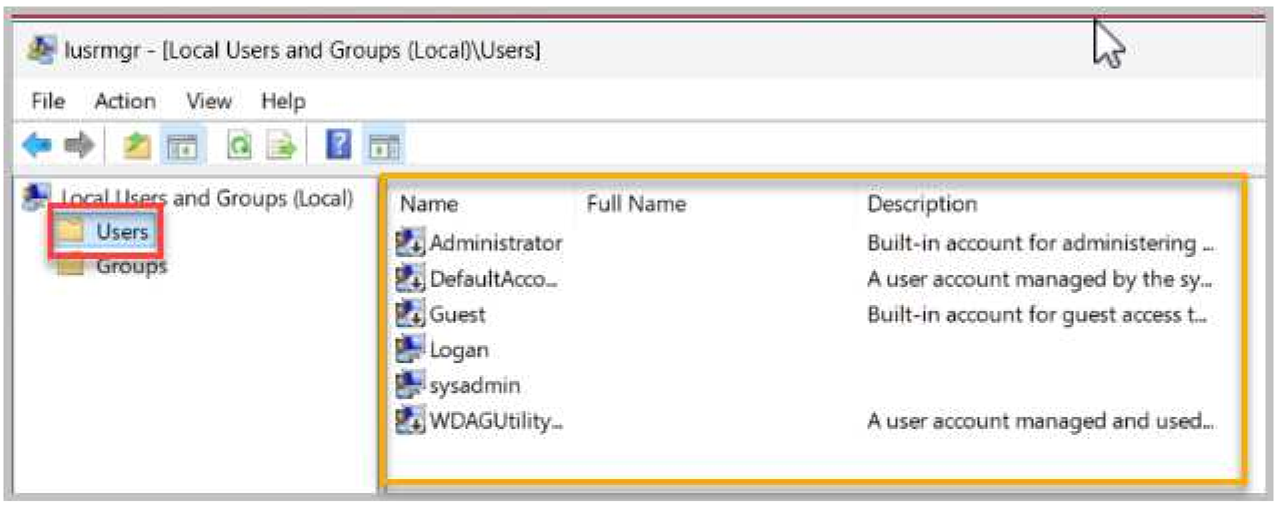
16. If prompted by the *User Account Control* window, select **Yes.**



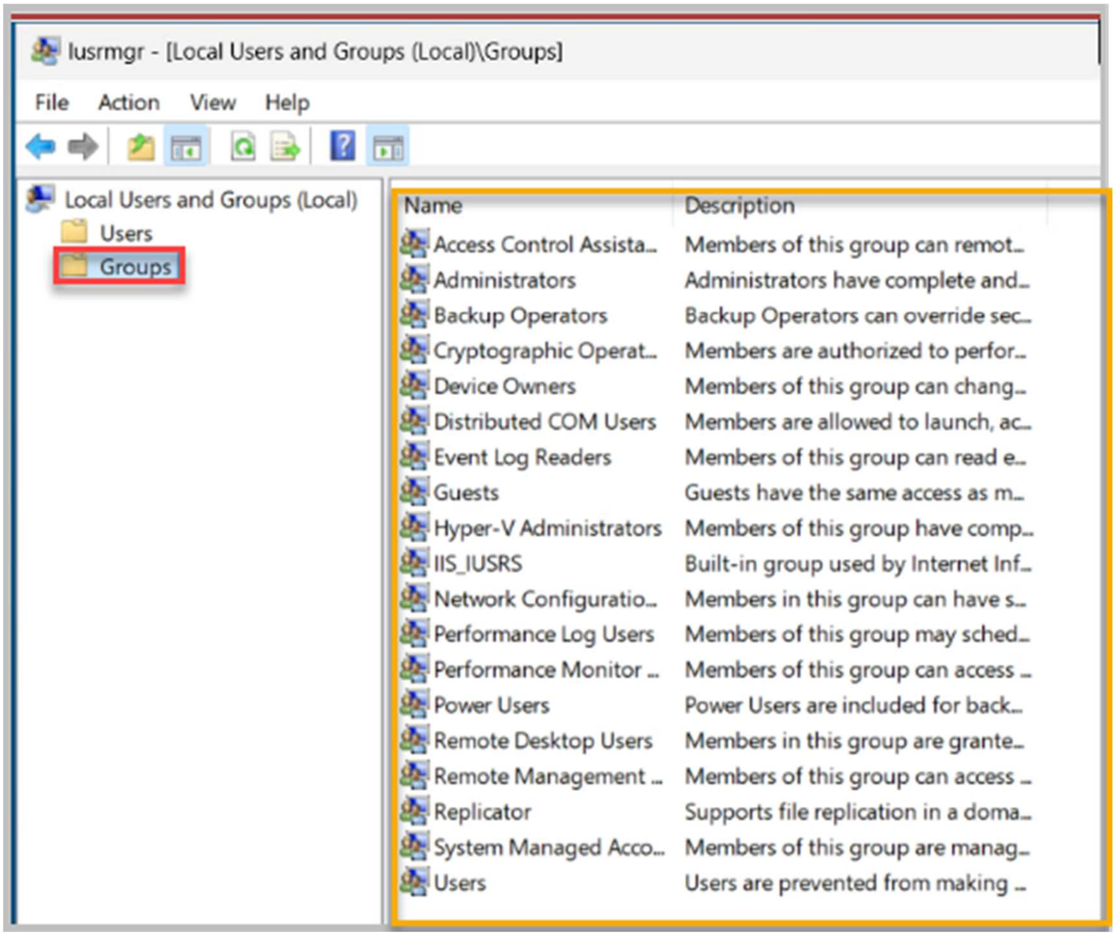
17. In the *Command Prompt*, type `lusrmgr.msc` and hit **enter** on your keyboard.



18. In the *lusrmgr – [Local Users and Groups (Local)\Users] window,* select **Users** first to see the full list of users that have access to the Windows 11 System.



19. Select **Groups** to see the full list of groups that have access to the Windows 11 System.



Please Note

- *(lusrmgr.msc)* is a Windows command that opens the "Local Users and Groups" management console, which is primarily used to manage user accounts and groups on a Windows computer.
- This tool allows administrators to add, remove, and manage user accounts and their permissions on the system.

20. Close all open windows and log out of the **sysadmin** account before moving to the next activity.

4 Active Directory Users and Computers in Windows 10

In this lab topology, the Windows client systems are configured within a Workgroup, which means they do not communicate with or connect to any Active Directory domain servers. This setup restricts the systems from accessing many domain-specific features, including centralized authentication, policy management, and various domain services typically provided by an Active Directory environment. Windows systems connected to a domain network, however, have access to an extensive suite of administrative capabilities through the Remote Server Administration Tools (RSAT). RSAT allows IT administrators to remotely manage server roles, features, and configurations on a Windows Server system directly from a Windows client, streamlining management of domain resources through a single interface.

RSAT can only be installed on Windows client operating systems running Professional, Enterprise, or Education editions, as it is not available for Home editions. Since the October 2018 update of Windows 10 (version 1809), Microsoft has made RSAT a "Feature on Demand," meaning that all Active Directory and RSAT tools are now components you can add directly through the Windows Settings interface without a separate download. This update ensures that tools like Active Directory Users and Computers are easily accessible and kept up to date alongside Windows updates, enhancing convenience and reducing compatibility issues.

Among the RSAT tools, one of the primary tools for managing Active Directory domains is the **Active Directory Users and Computers (ADUC)** MMC (Microsoft Management Console) snap-in. ADUC enables domain administrators to perform essential domain management tasks, such as creating and managing user accounts, setting up and organizing groups, managing computers within the domain, and structuring organizational units (OUs) that reflect a company's hierarchy or operational units. These organizational units allow for the efficient application of Group Policies and facilitate user and computer management within specific areas of the organization.

Additionally, ADUC provides tools to enforce policies, reset passwords, enable or disable accounts, manage group memberships, and assign permissions within the Active Directory. With ADUC, administrators can also control other aspects of Active Directory, such as performing advanced queries to locate specific objects and managing delegation to assign permissions to different administrators or IT staff.

This comprehensive approach to managing users, computers, and permissions makes ADUC a cornerstone of any organization's IT infrastructure, providing a framework for securing resources, organizing network objects, and enforcing compliance across a domain.

1. The lab is now complete; you may end the reservation.

